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Lower bounds for the problem of sequencing cutting patterns

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Abstract:

In some wood cutting settings, the cutting sequence of the patterns produced to minimize trim waste and satisfy customers' requirements is important. Each different piece size cut is put into a stack which is removed only after the last piece of that size is cut. It is desired to minimize the maximum number of stacks often because of space limitations and/or material handling problems. In this work we present some possible ways of computing lower bounds for this optimization problem. Lower bounds are of interest in general implicit exhaustive search procedures to solve this pattern sequencing problem. Limited computational results are presented to assess the quality of the bounds.

Keywords: cutting stock, pattern sequencing, pattern allocation, graph traversing

1. INTRODUCTION

The minimization of open stacks problem (MOSP) deals with the sequencing of the patterns in order to minimize the maximum number of incomplete stacks during the cutting run. A stack is opened for each different panel type cut and it is closed only after the last piece of that type is cut. Closed stacks can be removed prior to the cutting of the next pattern. The MOSP also arises in other production settings as described in Yanasse⁷.

In Yanasse^{6,7}, a non-trivial mathematical formulation, a branch and bound algorithm and a greedy heuristic algorithm are suggested. Relations of the MOSP with the minimization of order spread problem^{1,4} and the minimization of the number of tool switches problem⁵ are also discussed.

Lins³ proposed a heuristic method to solve the MOSP for the particular case where there are at most 2 different panel types per pattern and the graph associating the nodes to orders and arcs to patterns is a 1-tree, i.e. a tree with an additional arc.

Yanasse¹⁰ focused the case with at most two panel types in each pattern. He presented exact polynomial algorithms when the graph corresponding to the problem is a

tree, a 1-tree or a "0-1 common vertex polygon" graph. Yuen^{11,12,13} proposes several simple heuristics for pattern sequencing.

Yanasse⁸ proved that an optimal solution to *any* MOSP can be obtained by solving a corresponding MOSP where all patterns have at most two panel types. The corresponding MOSP can be obtained in polynomial computational time, hence, solving general MOSPs is as difficult as solving "simpler" ones where all patterns have at most two panel types.

Yanasse⁹ proposes a heuristic procedure which uses MOSP's equivalent graph representation. A conveniently selected subgraph of the original graph with a simpler topology is considered which can be solved exactly without difficulty. A solution to the original problem is constructed using as reference this simpler partial graph solution.

We are unaware of other published works where this same problem is considered. We are also unaware of any result with respect to the computational complexity of solving MOSP. MOSP close similarity with other NP-hard combinatorial problems² seems to indicate that this problem is computationally difficult to solve.

In this paper we discuss possible ways for determining lower bounds for the MOSP. Lower bounds are useful in branch-and-bound schemes for solving this problem and to assess the quality of solutions obtained by heuristics.

2. Lower bounds for MOSP

One trivial lower bound for MOSP, indicated in Yanasse^{6,7}, is the maximum number of different pieces in a pattern. We will denote the lower bound obtained in this way by LB^1 . Consider the instance of MOSP given in Table 1.

Table 1

Patterns	Order Parts
P ₁	a d g i
P ₂	d
P ₃	b e i j
P ₄	a f h j
P ₅	c j
P ₆	b f j
P ₇	d g h
P ₈	f
P ₉	a d
P ₁₀	e g i j

P₁ (also, P₃, P₄ or P₁₀) has 4 different order parts, therefore, the minimum number of stacks that will be opened in an optimal solution is 4, that is $LB^1 = 4$.

Another immediate lower bound⁷, which we will denote by LB^2 , can be obtained observing that whatever the solution of this sequencing problem is some stack will be completed first. Hence, a lower bound for MOSP can be obtained by comparing the number of open stacks obtained when each one of the pieces in the problem is completely cut and choosing the smallest value.

Consider again the instance of MOSP given in Table 1. Some stack corresponding to part, a, b, c, d, e, f, g, h, i or j must be completed first. To complete some part, all the patterns that contains it must be cut. For instance, patterns P_1 , P_4 and P_9 must be cut in order to finish part a, resulting in at least 7 open stacks, one corresponding to each different part cut. If we compute the resulting number of stacks for each one of the pieces in the problem, the smallest value obtained is 2, which is obtained by completing piece c. Therefore, we obtain a lower bound $LB^2 = 2$ for the optimal solution to MOSP.

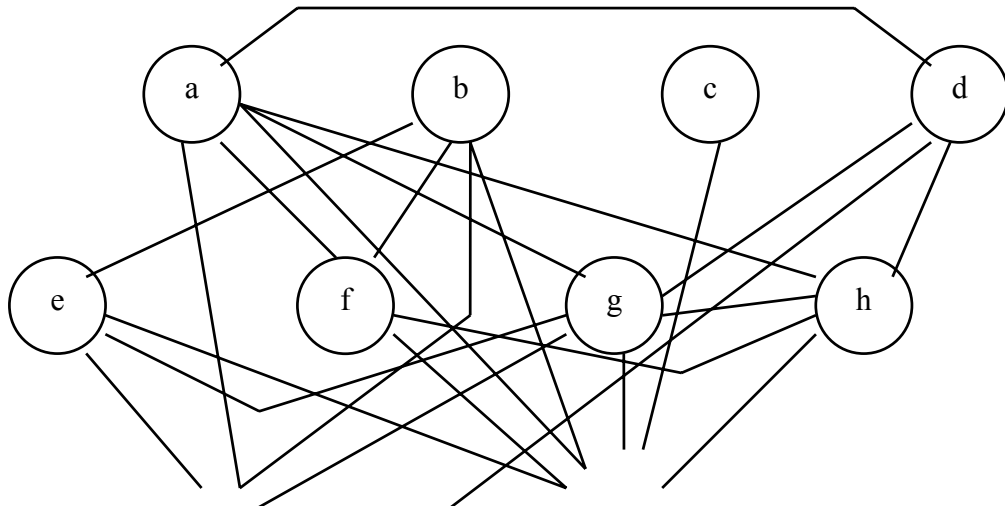
Yanasse⁸ has identified some solution properties of the MOSP that can be useful in determining other lower bounds:

Proposition 1 (reproduced from Yanasse⁸): Suppose we have a MOSP, $PROB_1$, with N patterns to be sequenced. Suppose we have a second problem, $PROB_2$, with the same N patterns as problem $PROB_1$ plus a few more patterns. Then the optimal solution value of $PROB_2$ is at least the optimal solution value of $PROB_1$.

Proposition 2 (reproduced from Yanasse⁸): Let $G(V,E)$, the graph obtained from a problem $PROB$, be a connected non-empty graph where all nodes have the valency n ($n \geq 2$) or more. Then, a lower bound on the optimal solution value to $PROB$ is $n+1$, that is, in an optimal solution to $PROB$, we must have at least $n+1$ open stacks at some time during cutting.

Using Proposition 1 one can obtain a lower bound for MOSP by deleting arcs of the original graph representation of the problem. Once a simpler graph that can be solved exactly is obtained, its solution value can serve as a lower bound for the original problem. We will denote a lower bound obtained this way as LB^3 .

Consider again the instance of MOSP given in Table 1. The graph representation of this data set is presented in Figure 1.



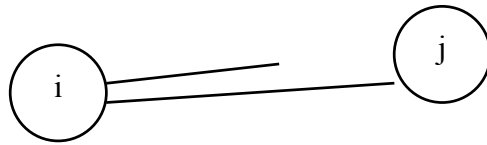


Figure 1

By using proposition 1, if we delete all arcs incident to nodes c. The resulting graph is presented in Figure 2.

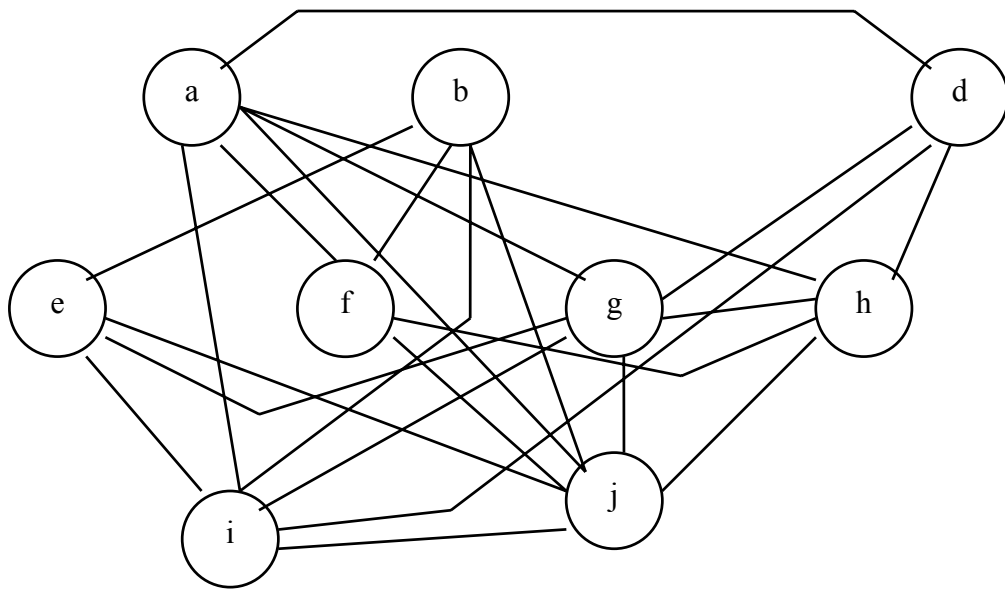
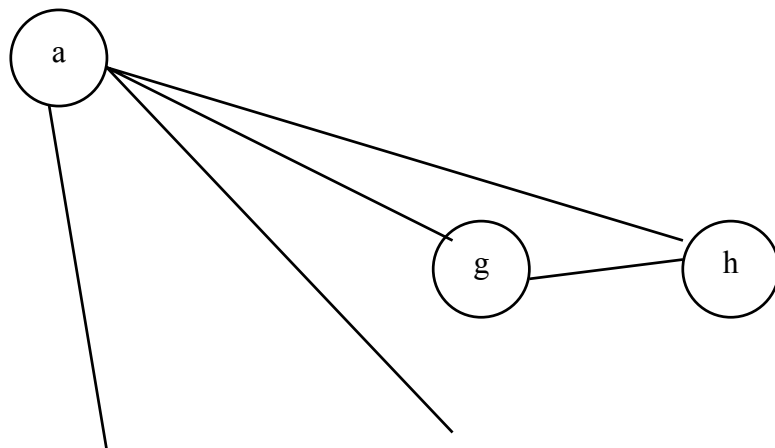


Figure 2

The problem seems still complex to solve. If we delete all arcs incident to nodes b, d, e and f which are the nodes in the graph with the smallest degree, the resulting graph is presented in Figure 3.



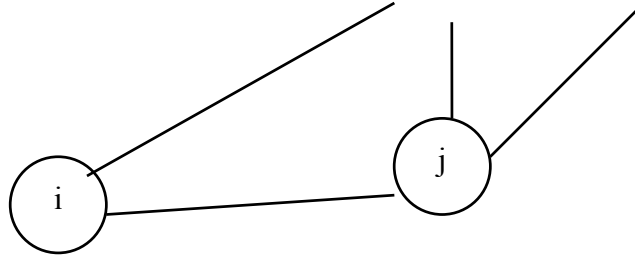


Figure 3

The solution can be obtained without much difficulty with resulting solution value of 4 open stacks. This is, hence, a lower bound on the optimal solution to the original problem, that is, $LB^3 = 4$.

Proposition 2 can be used directly by analysing the degree of the nodes of the graph representation of the problem. The lower bound obtained this way will be denoted LB^4 . In the graph of Figure 1, all nodes have degree at least 1, hence, a lower bound on the optimal solution value is 2 ($LB^4 = 2$). Observe that, due to proposition 1, every subgraph of the original graph can also be considered in this analysis. So, in the graph of Figure 1, we could attempt to delete node c and all the arcs incident to it, since it is the node with the smallest degree. In the resulting subgraph (Figure 2) the least degree is 4, hence, a lower bound value of 5 is attained. No better lower bound value is obtained by deleting more arcs in the subgraph of Figure 2.

We present next, a new observation which may serve as well to determine a lower bound for MOSP: The lower bound obtained this way will be denoted LB^5 .

Proposition 3: Let $G_1(V_1, A_1)$ be the corresponding graph of a MOSP $PROB_1$. Let $G_2(V_2, A_2)$ be the hypergraph obtained from $G_1(V_1, A_1)$ by collapsing a single arc $a \in A_1$. As the result of this arc collapse, a single supervertex v is obtained in place of the vertices $v_1 \in V_1$ and $v_2 \in V_1$ incident to arc a . Let's denote $PROB_2$ the surrogate MOSP corresponding to such hypergraph $G_2(V_2, A_2)$. Let SEQ_1 be any sequencing of the patterns in problem $PROB_1$. Let SEQ_2 be exactly as SEQ_1 excluding the pattern corresponding to arc a in graph $G_1(V_1, A_1)$. SEQ_2 is, hence, a feasible solution to $PROB_2$. If we follow these sequences, after each corresponding pattern sequenced cut, the number of open stacks resulting using SEQ_1 in problem $PROB_1$ is at least as large as the number of open stacks resulting using SEQ_2 in problem $PROB_2$.

Proof of Proposition 3: The open stacks using SEQ_1 and SEQ_2 are the same until the first arc incident to node v_1 or v_2 is sequenced. Let's assume without loss of generality that the node v_1 is the first to be opened. In SEQ_2 , in correspondence to this arc, supernode v will be opened. If in SEQ_1 an arc is sequenced such that v_2 is opened, in SEQ_2 , no new node is opened. In particular, if this arc in SEQ_1 is a , v_1 may or may not be closed. Anyway, after each corresponding pattern sequenced, the number of open stacks using SEQ_1 will be at least as large as the number of open stacks using SEQ_2 .

As a direct corollary of Proposition 3, we have:

Corollary 1: The optimal value of PROB_2 is a lower bound for the optimal value of PROB_1 .

Consider the graph representation of the problem given in Figure 1. We are interested in stronger lower bounds, hence, we will try to collapse arcs incidents to nodes with the smallest degree value vertices. Since vertex c is the one with the smallest degree value, we may collapse the arc that incident to it. The resulting hypergraph is presented in Figure 2.

No improvement of bounds (which at this moment is 5 by computing LB^4 in this graph) has been achieved at this time after collapsing the first arc.

Consider now the graph in Figure 2. As we described previously, we are interested in stronger lower bounds, hence, we will try to collapse arcs incidents to nodes with the smallest degree value vertices. Since vertices b , d , e and f are the ones with the smallest degree value, we may collapse, for instance, the arc that is incident to b and e . The resulting hypergraph is presented in Figure 4.

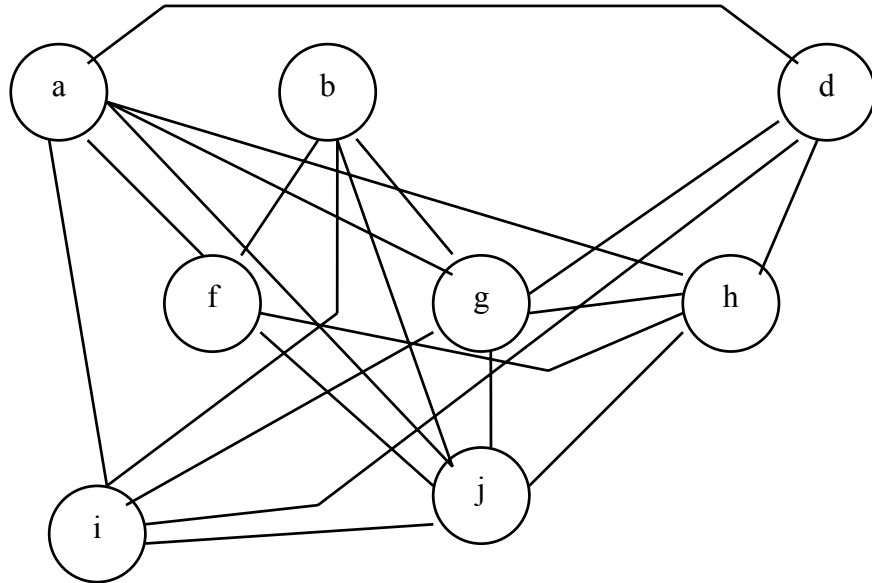
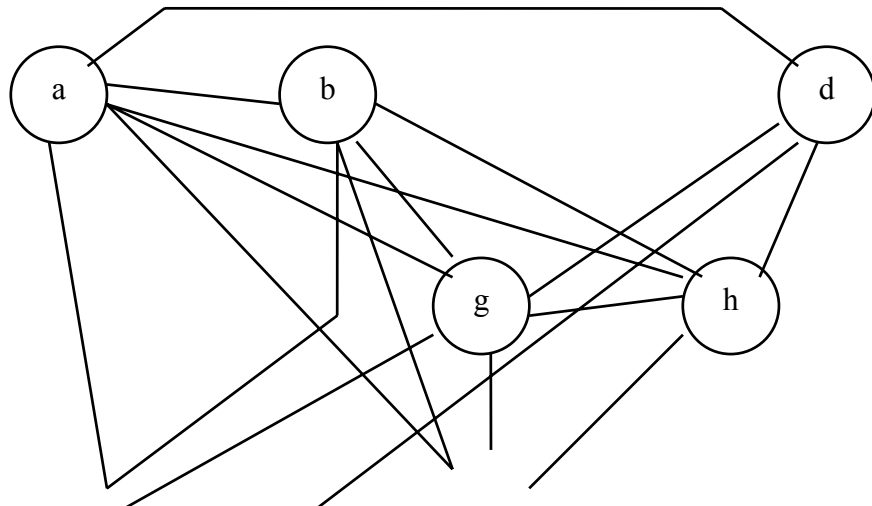


Figure 4



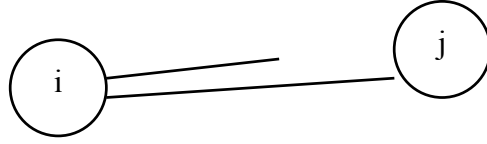


Figure 5

Consider now the graph in Figure 4. No improvement of the bound is obtained. We proceed in the same way and we may collapse, for instance, the arc that is incident to b and f. The resulting hypergraph is presented in Figure 5.

Consider now the graph in Figure 5. Again, no improvement of the bound is obtained. We proceed in the same way and we may collapse, for instance, the arc that is incident to d and h. The resulting hypergraph is presented in Figure 6.

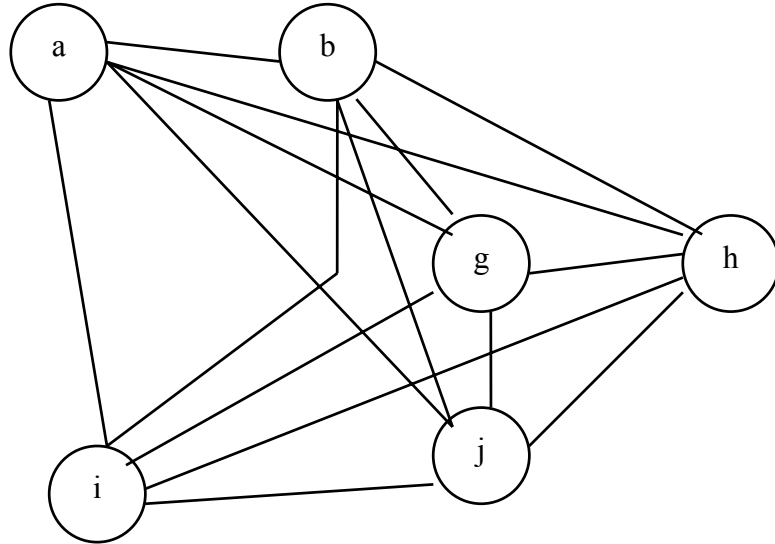


Figure 6

The resulting graph in Figure 6 is complete. Therefore $LB^5 = 6$, for this instance. Notice that we could obtain this bound using LB^4 in this graph since all nodes have degree at least 5.

As observed in the instance presented, the collapsing procedure produces stronger lower bounds compared with the simple arc deletion procedure proposed previously. This happens because the collapsing procedure may increase the degree of some of the remaining nodes of the graph. With the deletion procedure this never happens.

We formalize next the proposed general collapsing procedure for determining lower bounds for the MOSP. We assume the MOSP is given in its corresponding graph problem and, without loss of generality, that the graph is connected.

Procedure LBArcCollapse

- Step 0 - Compute $LB = \max \{LB^1, LB^2, LB^3\}$, the initial lower bound for the problem and M , the number of nodes in the graph considered.
- Step 1 - Compute the degree of each one of the M nodes in the graph. Compute LB^4 . If $LB^4 > LB$, then $LB = LB^4$. If $N \leq LB+1$, then $LB^5 = LB$ and STOP. Else, let S be the set of nodes with the smallest degree. Choose any arc, say a , in the graph which is incident to two nodes whose sum of their degrees is the minimum and such that, at least one of the nodes belongs to set S .
- Step 2 - Collapse arc a , and as the result of this arc collapse, a single supervertex is obtained in place of the two vertices incident to a . All the arcs that were incident to these two vertices are now incident to the supervertex; in case of duplication of arcs, maintain only a single one. Set $M \leftarrow M-1$ and return to Step 1.

We present next some computational tests comparing all these lower bounds.

3. Computational experiments

To compare the lower bounds proposed, we performed some tests on randomly generated instances of MOSP. They were generated with the following characteristics:

N = total number of patterns;

M = total number of different panel types;

C = maximum number of different panel types in each pattern.

For each pattern, we consider that the number of different panel types is uniform between $[1, C]$.

The following cases were considered:

i. $N=10, M=10, C=4$

ii. $N=20, M=15, C=8$

iii. $N=30, M=25, C=10$

iv. $N=40, M=30, C=15$

In Table 2 the results obtained are presented using ?? instances for each case considered.

Testes

A tabela a seguir, mostra o resultados dos testes comparativos efetuados para analisar o desempenho dos *Lower Bounds*. Na coluna “Caso de teste” está indicado o

nome do arquivo de dados utilizado. Nas colunas LB1, LB2, LB3, LB4 e LB5 estão indicados os valores dos *Lower Bounds* obtidos para o caso de teste indicado. Na coluna “n peças” está indicado a quantidade de peças considerada (número de colunas geradas). Na coluna “n padrões” está indicada a quantidade de padrões considerada (número de linhas). Na coluna “C” está indicado a quantidade máxima de peças que um padrão pode conter (quando não explicitado, é porque o caso de teste foi gerado manualmente).

Caso de teste	LB1	LB2	LB 3	LB4	LB5	n peças	n padrões	C
Tabela1	3	2	3	3	3	6	5	
pattern0	04	04	04	04	04	06	04	
pattern2	04	02	04	04	04	16	14	
pattern4	04	02	04	04	04	20	18	
pattern5	04	02	03	04	04	20	18	
pattern6	04	02	04	04	04	18	16	
pattern7	03	03	03	03	03	06	03	
pattern8	02	02	03	04	04	16	22	
pattern9	02	02	03	05	05	18	33	
pattern10	02	02	05	05	05	18	35	
pattern11	04	02	03	06	06	20	19	
Caso101	04	04	04	06	06	10	10	04
Caso102	04	02	03	05	05	10	10	04
Caso103	04	03	03	05	05	10	10	04
Caso104	04	03	03	06	06	10	10	04
Caso201	08	05	05	11	11	20	15	08
Caso202	08	08	08	14	14	20	15	08
Caso203	08	03	03	13	13	20	15	08
Caso204	08	05	05	12	12	20	15	08
Caso301	10	04	04	19	19	30	25	10
Caso302	10	10	10	17	18	30	25	10
Caso303	10	05	05	18	19	30	25	10
Caso304	10	11	11	20	21	30	25	10
Caso401	15	23	23	30	32	40	30	16
Caso402	15	12	12	26	27	40	30	16
Caso403	15	22	22	27	29	40	30	16
Caso404	15	17	17	30	32	40	30	16
Caso501	02	02	03	03	03	10	10	02
Caso502	02	02	03	00	03	10	10	02
Caso503	02	02	03	03	03	10	10	02
Caso504	07	02	04	07	07	10	10	02
Caso601	05	02	03	07	07	30	30	04
Caso602	04	03	03	06	07	30	30	04
Caso603	04	02	03	07	07	30	30	04
Caso604	05	02	03	06	07	30	30	04
Caso701	08	06	06	17	18	40	40	08
Caso702	08	03	03	14	15	40	40	08
Caso703	08	07	07	18	19	40	40	08
Caso704	08	04	04	16	18	40	40	08
Caso801	12	13	13	25	27	50	50	12
Caso802	12	18	18	30	32	50	50	12

Caso803	12	16	16	25	28	50	50	12
Caso804	12	15	15	28	31	50	50	12
Caso901	12	15	15	29	32	55	55	12
Caso902	12	05	05	28	30	55	55	12
Caso903	12	10	10	29	31	55	55	12
Caso904	12	14	14	29	32	55	55	12
Caso905	12	07	07	23	25	55	55	12
Caso906	12	23	23	30	33	55	55	12
Caso907	12	06	06	28	30	55	55	12
Caso908	12	06	06	27	29	55	55	12
CasoA01	12	13	13	27	30	60	60	12
CasoA02	12	14	14	28	31	60	60	12
CasoA03	12	15	15	31	35	60	60	12
CasoA04	12	20	20	32	35	60	60	12
CasoB01	12	08	08	29	32	70	70	12
CasoB02	12	15	15	29	32	70	70	12
CasoB03	12	08	08	28	31	70	70	12
CasoB04	12	24	24	32	35	70	70	12
CasoC01	12	17	17	36	40	80	80	12
CasoC02	12	15	15	31	34	80	80	12
CasoC03	12	11	11	33	37	80	80	12
CasoC04	12	13	13	37	41	80	80	12
CasoD01	12	17	17	36	40	90	90	12
CasoD02	12	16	16	34	39	90	90	12
CasoD03	12	08	08	34	37	90	90	12
CasoD04	12	03	03	35	39	90	90	12
CasoE01	24	45	45	78	82	100	100	24
CasoE02	24	65	65	73	77	100	100	24
CasoE03	24	53	53	68	72	100	100	24
CasoE04	24	67	67	76	80	100	100	24
CasoF01	24	45	45	83	91	150	150	24
CasoF02	24	69	69	88	95	150	150	24
CasoF03	24	69	69	93	101	150	150	24
CasoF04	24	54	54	92	100	150	150	24

Tabela 1: comparação entre os *Lower Bounds*

3. SUMMARY AND CONCLUSIONS

Different possible ways of calculating lower bounds for the minimization of open stacks problem are discussed. A new promising procedure for determining a lower bound for MOSP was introduced. It uses the graph representation of the problem followed by a arc collapsing procedure that reduces the graph in an attempt to make easier the analysis. The new procedure, as expected, found stronger bounds on all instances tested.

The procedure presented is quite simple and can be used to generate a good lower bound for the problem. It might be helpfull to use in some branch and bound scheme. The arc-collapsing scheme was used in Yanasse?? where a heuristic procedure was proposed which presented god computational results.

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